Willett Engineering Company
Consulting Structural Engineers

3528 Habersham at Northlake
Tucker, Georgia 30084
Phone: (770) 270-9484
www.WillettEngineering.com

STRUCTURAL OBSERVATION REPORT

DATE: August 29, 2025

JOB NUMBER: 225212

JOB NAME: Historical Church
202 Broad St, Griffin, GA

ATTENTION: Bruce Ballard, Griffin Downtown Development Authority

FROM: J. Mac Willett, P.E., S.E.

Please note that the observations listed below are based on what was visible to the Engineer during the inspection. Certain elements of the building may have been hidden from view, and as such, this report does not constitute a comprehensive inspection of the entire structure. There may be aspects of the structure that were not observed or could not be assessed during this visit.

1. Introduction

As requested, I conducted a site visit to a historical church located at 202 Broad Street in Griffin, GA. The purpose of this visit was to observe and assess the condition of the existing structure. The building, approximately 3,800 square feet in area, shows signs of neglect and appears vacant of occupants. The building exhibits clear signs of neglect, including graffiti, trash, lack of rain leaders, and boarded-up openings.

2. Building Description

The church is a two-level structure comprised of an administrative lower level (slab-on-grade), an elevated sanctuary level (wood framed), and a roof (wood gable trusses).

The building is constructed with loadbearing clay masonry walls and framed with 2x8 wooden floor joists. The floor framing spans 14 feet between 6x6 wooden beams, which are supported by 6x8 wooden posts spaced 6 feet on center. These posts appear to be founded directly on the concrete slab. Roof trusses clear span approximately 40 feet and are spaced at roughly 14 feet on center.

3. Modifications and Alterations

Over the years, the building has undergone several modifications and renovations. I was able to observe the following suspected changes from the original building:

- Removal of the choir loft.
- Installation of a lay-in ceiling to conceal mechanical ductwork.
- Addition of openings in the exterior masonry walls.
- Installation of a new concrete masonry knee wall adjacent to the exterior wall on the lower level.

4. Structural Analysis, Code Compliance, and Safety Considerations

Southern Yellow Pine is the common wood species used in the region. Because older wood is generally stronger than newer wood in terms of allowable stresses, graded #1 Southern Yellow Pine lumber was assumed for analysis purposes. My calculations indicate that the wood floor framing is adequate to support a design live load of approximately 50 pounds per square foot. It should be noted this is less than the current code-mandated live load for use as a sanctuary, which is 100 pounds per square foot in the

current Georgia building codes.

Direct observation of the roof framing was not possible. The structure's longevity suggests that the roof load capacity is sufficient, and no significant distress was observed. This, however, does not eliminate the possibility of structural issues concealed beneath the roofing. It is recommended that the shingles be removed and additional inspection be conducted prior to any restoration or renovation work. Personnel should also be made aware of this risk prior to performing any roof top work and should take necessary precautions.

The exterior loadbearing masonry walls consist of three wythes of brick, typical of historical masonry construction. These walls extend approximately 25 feet from the foundation to the eave and 32 feet at the ridge, supporting the roof framing. Although this was common practice at the time of construction, the masonry walls would not be able to meet current code-required load demands without reinforcement. Deterioration of mortar would also compromise the walls' structural integrity.

As a historical structure, the building does not comply with current building codes or practices. Additional observed deficiencies include:

- Absence of mechanical anchors for wood-to-wood and wood-to-masonry connections. Positive attachment is required by current code.
- Unreinforced masonry walls. Reinforced masonry is required for this classification of building in current code.
- No lateral load path was observed. Historically, structures were not designed to withstand wind or seismic events. It is unclear if this structure would be able to resist these loads should the building experience high winds or an earthquake.

5. Observations of Deterioration

Signs of neglect and deterioration include:

- Water intrusion, particularly evident in the sanctuary ceiling.
- Moisture on the interior side of exterior walls.
- Absence of rain leaders, causing water discharge at the base of walls and increasing the risk of localized scour and settlement.
- Termite damage in at least one wood column.
- Removal of a wooden post intended to support a 6x6 beam.
- Compromised wooden stairs between levels, deemed unsafe for use.
- The mortar between bricks in several areas appears sandy, indicating compromised bonding capacity.
- Cracking was observed above some exterior wall openings.
- Exposed wood elements may be deteriorated due to rot.
- Modifications to the masonry walls appear to have been performed without the guidance of licensed engineers or contractors, and may not be structurally sound.

6. Conclusion and Recommendations

In its current state, it is my professional opinion that the building should not be occupied. However, as this structure has historical significance, a rehabilitation is feasible if the structure is intended to be occupied. A comprehensive rehabilitation plan should be developed, including:

- Moisture mitigation and drainage improvements.
- Structural reinforcement and repair.
- Restoration of historical architectural elements.
- Compliance upgrades where feasible.

A multi-disciplinary effort would be required for the complete rehabilitation of the structure. It should be emphasized that firms with experience in historical preservation may be best suited to preserve the historical value of the structure while also providing the necessary upgrades and reinforcement to safeguard occupants and the public.

Thank you for contacting Willett Engineering Company. Please let us know if we can be of further assistance on this project.

Broad Street View

Photo 1

Overhang
Deflection
Openings
Cut Into
Exterior Wall
Eroded
Mortar Joints
Broad Street

Photo 2

Exposed
Wood
Deterioration
Repaired/
Replaced
Masonry
Cracking
In
Masonry

Photo 3

Deflection at Overhang
Missing Rain Leader
Missing Rain Leader
Photo 4

Ceiling Damage (water)
NE Corner of the Sanctuary

Photo 5

Lay-In Ceiling Grid
Water Damage on The South Side of the Roof

Photo 6

Repaired Brick Wall

Photo 7

Moisture Damage on Exterior Wall - Facing Broad Street

Photo 8

2x8 Wooden Joist
6x6 Wooden Beams
Termite Damage in 6"x8" Wood Post
Photo 9

6x8 Wooden Post Bearing on Concrete Slab
Photo 10

Photo 11

Photo 12

Concrete Block Knee Wall
Recessed Concrete Slab
Photo 13

Wooden Lintel
Loose Laid Masonry
Photo 14

Wall Damage - Results of Wall Modifications

Wall Pockets for Previous Choir Loft Framing

Photo 15

Floor Framing into Loadbearing Brick Wall
Photo 16